



09 Database / PostgreSQL

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Agenda

- What is a database?
- Types of databases: Relational vs. Non-relational
- MySQL, PostgreSQL, MongoDB
- Installing and setting up PostgreSQL
- SQL Fundamental
 - DDL
 - CRUD
 - Filtering
 - Aggregate Function
 - Joins

What is Database

- A database is an organized collection of data that can be easily accessed, managed, and updated.
- Used to store information like user profiles, product inventories, transactions, etc.
- Enables efficient data retrieval and manipulation

Types of databases: Relational vs. Non-relational

- Relational Databases (RDBMS):
 - Structured data stored in tables
 - Uses SQL for querying
 - Enforces relationships via keys
 - Example: MySQL, MS SQL Server, PostgreSQL
- Non-Relational Databases (NoSQL):
 - Flexible schema (documents, key-value, graph, etc.)
 - Designed for scalability and unstructured data
 - Often used in real-time applications
 - Example: MongoDB, Cassandra, Redis
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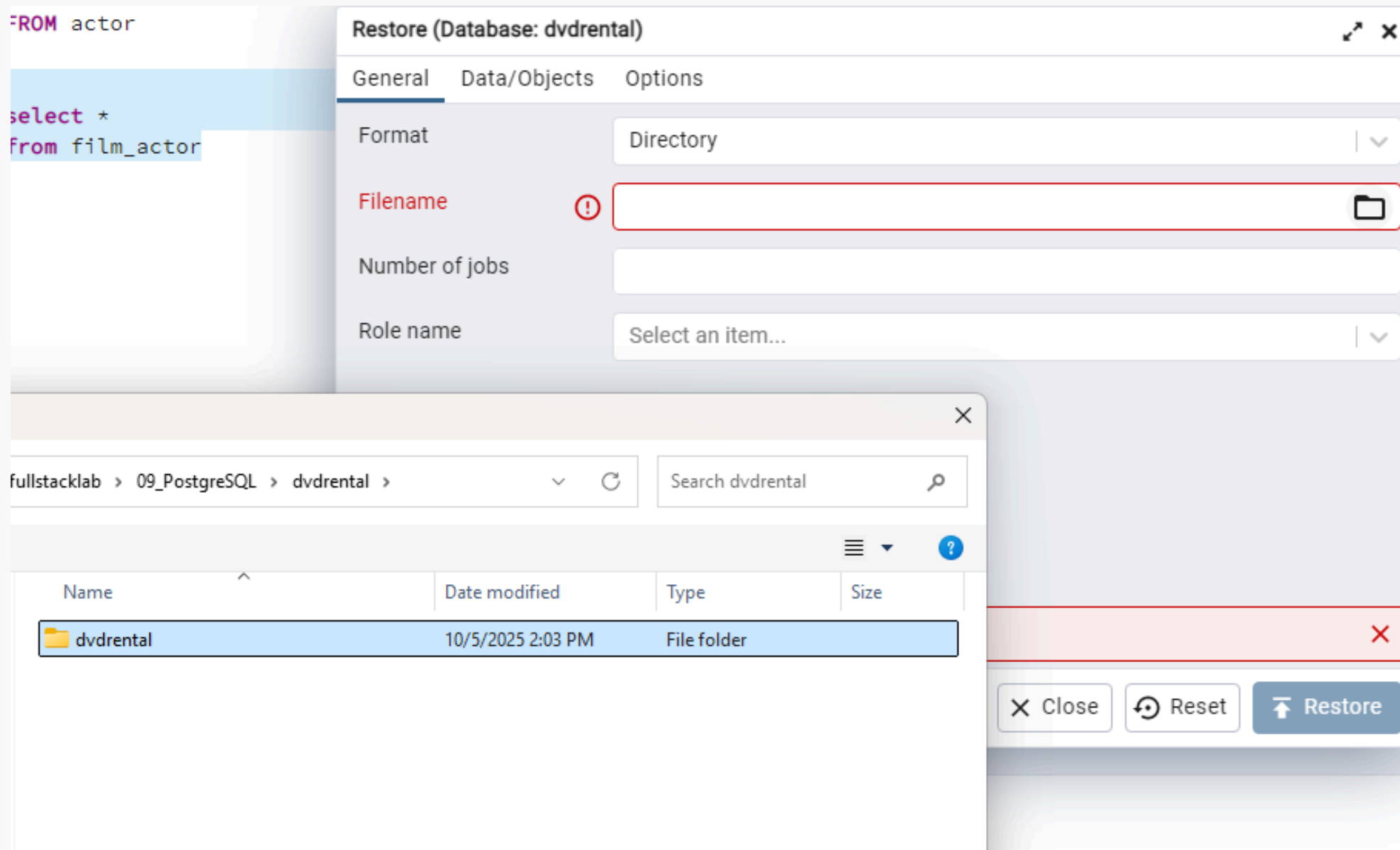
Installation and setting up PostgreSQL

- <https://www.postgresql.org/>
- Let's create a new database

SQL Fundamental: DDL

- Create
- Alter
- Drop

Restore sample database



SQL Fundamental: CRUD Operation

- Insert
- Select
- Update
- Delete

SQL Fundamental: Filtering Data

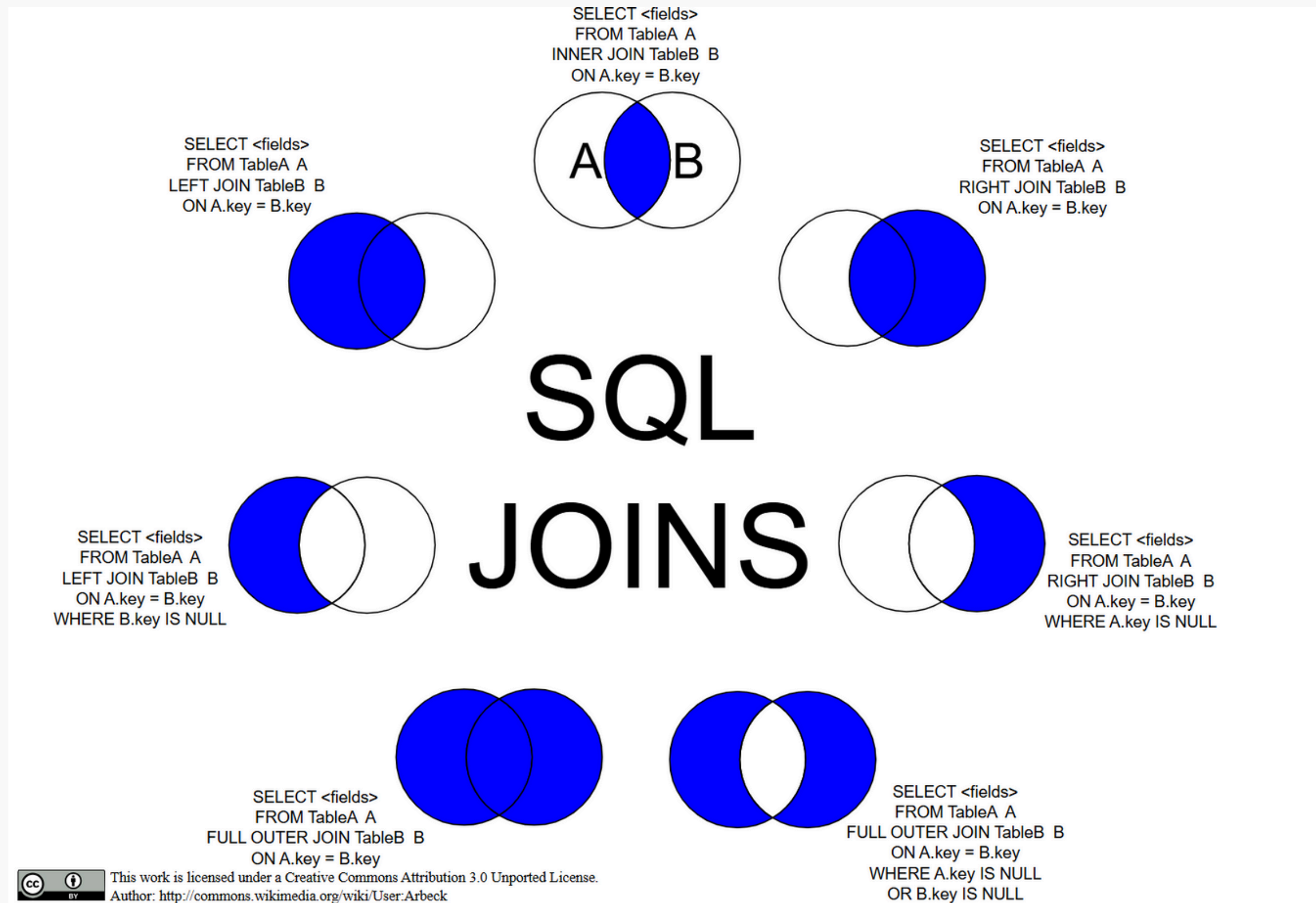
- Where

SQL Fundamental: Aggregate Function

- Count
- Sum
- Average
- Min
- Max
- Filter with Having

SQL Fundamental: Joining

- Inner Join
- Left Join
- Right Join
- Full Outer Join



Assignments

- List all films released in the year 2006.
- Find all actors whose last name starts with “S”.
- Get a list of films along with their categories (film title + category name).
- Find all films (title) in the “Children” category.
- For each actor, list how many films they have acted in.

(Include actors with zero films or only those with >0 — specify)